# Ensuring Full Reproducibility at MedInsure

Based on MedInsure Health Claim Approval Case Study

# Problem Statement

Reproducibility Problem Identified at MedInsure

- During an audit, the team failed to reproduce the model's results.
- The original training run details were not documented or tracked.
- Missing information:
  - Data version used
  - Environment & library versions
  - Hyperparameter configuration
- This creates risks in:
  - Audit compliance
  - Model reliability
  - Trust in automated claim approvals

# Why Reproducibility Matters

- Required for auditability
- Ensures trust in model decisions
- Enables debugging & model comparison
- Essential for regulatory compliance (especially for insurance/finance)
- Prevents inconsistent performance across environments

# Key Components of Reproducibility

Data Versioning

Environment Metadata Tracking

Hyperparameter & Experiment Tracking

# 1. Data Versioning

◆ WHAT

- Store snapshots of datasets used for training
- Track schema + data transformations
- Assign unique version IDs/hashes

◆ WHY

- Training data changes constantly (ETL updates, missing features, new codes)
- Need to recreate exact dataset used for Model v1, v2, v3…

# Tools for Data Versioning

- DVC (Data Version Control)
- LakeFS
- Delta Lake Time Travel
- Git-LFS for large CSVs
- Cloud Storage Versioning (S3, Azure, GCS)

# 2. Environment Metadata

**Problems in case study:**

- Python 3.8 vs 3.10 mismatch
- Different scikit-learn, pandas versions
- Results differ between dev + staging

**What to store:**

- Python version
- Library versions (pip freeze)
- OS, hardware details
- Dockerfile or Conda environment

# Tools For Environment Tracking

- Docker containers
- Conda environment YAML
- Requirements.txt
- MLflow environment logging
- GitHub Actions build logs
- requirements.txt

# 3. Hyperparameter & Config Tracking

WHY

- Model accuracy depends on exact settings
- Need to recreate:

1. Learning rate
2. Max depth
3. Seeds
4. Preprocessing rules
5. Feature engineering logic

In case study:

Team didn’t save hyperparameters → cannot rerun training

# Tools for Hyperparameter Tracking

- MLflow Tracking
- Weights & Biases (W&B)
- Neptune AI
- YAML config files in Git
- Automated experiment IDs

# What MedInsure should implement–

- ✓ Data version
- ✓ Environment version
- ✓ Hyperparams
- ✓ Registered model artifact
- ✓ Linked experiment ID in MLflow

# Thank You